

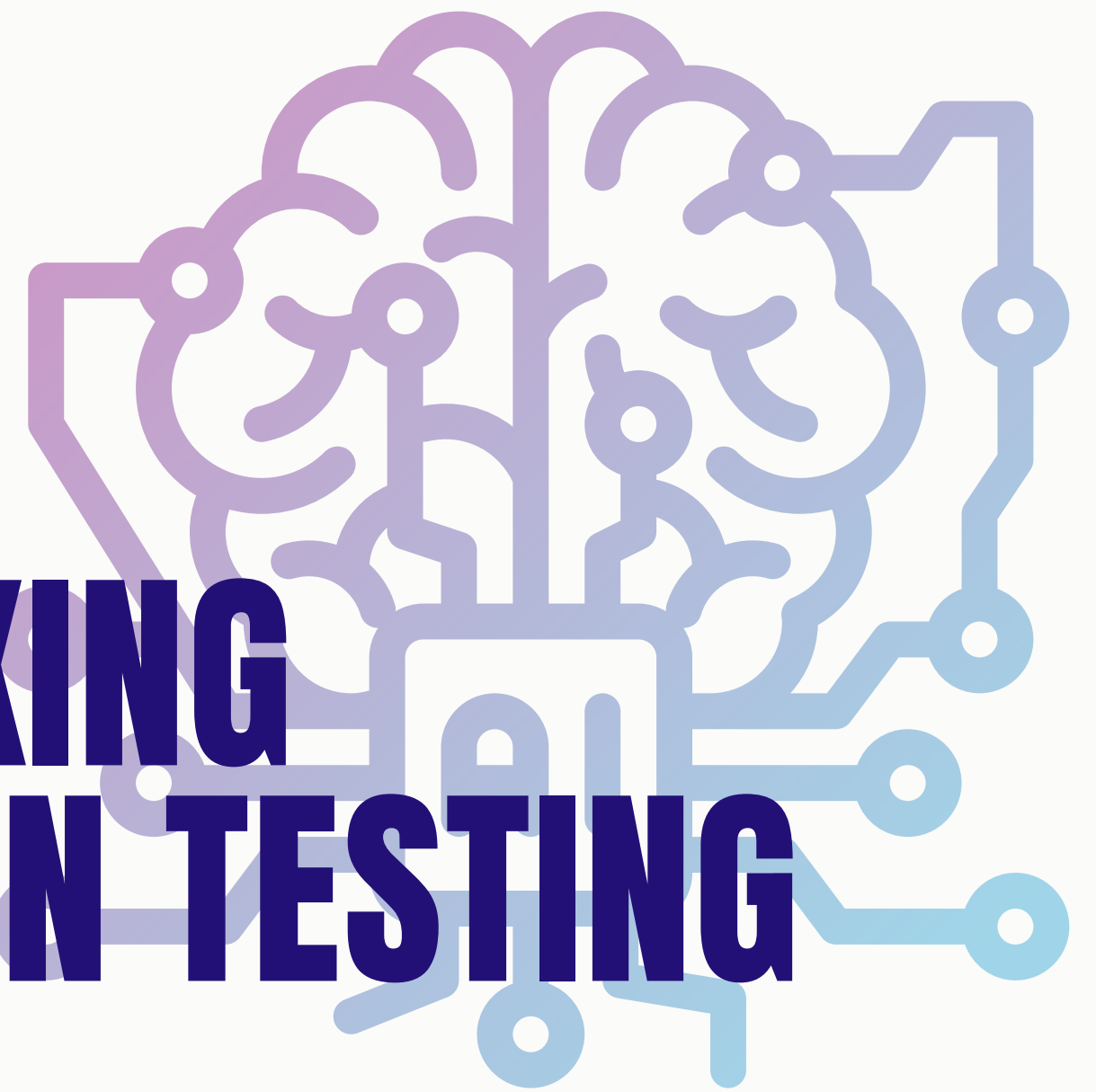
DIPLOMA PROJECT

DEVELOPMENT OF AI HACKING CHATBOT FOR PENETRATION TESTING

by Bagila Zhussupova, Akdidar Meldesh, Dana Artykova
CS-2220

**Department of Intelligent Systems and Cybersecurity
Astana IT University
6B06301 – Cybersecurity**

**Supervisor: Nurpeis Kulmuratov
Senior Lecturer**



PURPOSE

To develop an AI-powered Telegram chatbot that automates penetration testing and code vulnerability analysis of web applications, delivering real-time feedback and actionable security recommendations.

OBJECTIVES

- Analysis of existing AI-based penetration testing methods.
- Development of a chatbot with integrated pentest tools and AI model.
- Automation of code vulnerability detection and report generation.
- Implementation of ethical safeguards.
- Evaluation of system performance through testing and user feedback.

RELEVANCE

- Given that 70% of cyberattacks target web interface vulnerabilities and the high average cost of data breaches (\$4.45 million per incident, IBM 2023), there's a critical need for automated, ethical tools to improve security and raise awareness.

NOVELTY

- The bot offers full automation of penetration testing, AI-driven report generation, ethical safeguards, and an intuitive user-friendly interface.

OWASP ATTACKS DETECTION

- A01: Broken Access Control (via SSRF module, Subfinder, Gobuster, Fetch)
- A03: Injection (SQL Injection, Command Injection, XSS) (via SQLMap, XSSStrike, AI code analysis)
- A05: Security Misconfiguration (via Nikto, Nuclei, Gobuster, Fetch, WAF detection)
- A06: Vulnerable & Outdated Components (via Nikto)
- A08: Software and Data Integrity Failures (via Nikto – outdated software detection)
- A10: Server-Side Request Forgery (via custom SSRF testing module)

LITERATURE REVIEW

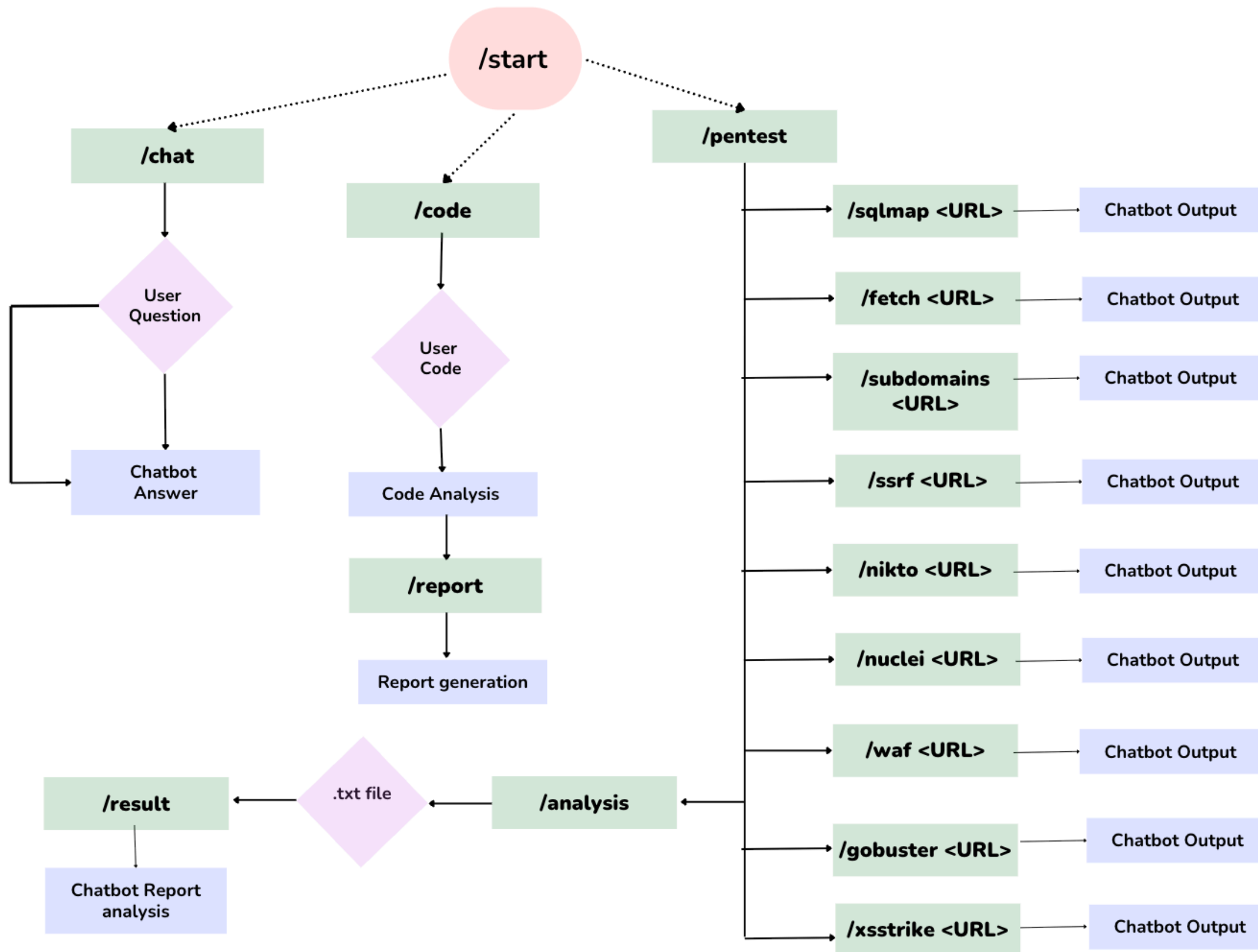
- **AI in Pentesting:** Recent research explores integrating LLMs (ChatGPT, GPT-4) to automate and enhance penetration testing workflows (Casey & Chamberlain, 2024; Hilario et al., 2024).
- **Tool Performance:** Studies confirm the efficiency of tools like SQLMap, Nikto, Nuclei, and Subfinder in identifying vulnerabilities (Vadla et al., 2024; Johari et al., 2024).
- **LLM Challenges:** Common issues include false positives, contextual limitations, ethical risks, and heavy computational needs (Gallus et al., 2023; Ozturk et al., 2023).
- **Telegram Bots:** Prior bot-based systems were lightweight but lacked real-time output, tool integration, and interactive reporting (Hafiz et al., 2023; Avisyah et al., 2023).
- **Educational Use:** AI-driven platforms support learning in offensive security but face challenges in real-world generalization (Alaryani et al., 2024; Hassanin & Moustafa, 2024).
- **ML for Detection:** ML models enhance SQLi detection and attack simulation but struggle with interpretability (Gudimetla, 2024; Akinsola et al., 2020).
- **Strategic Use:** ChatGPT aids in summarizing vulnerabilities for professionals, but overreliance may reduce critical thinking (Prasad et al., 2023; Francois et al., 2024).



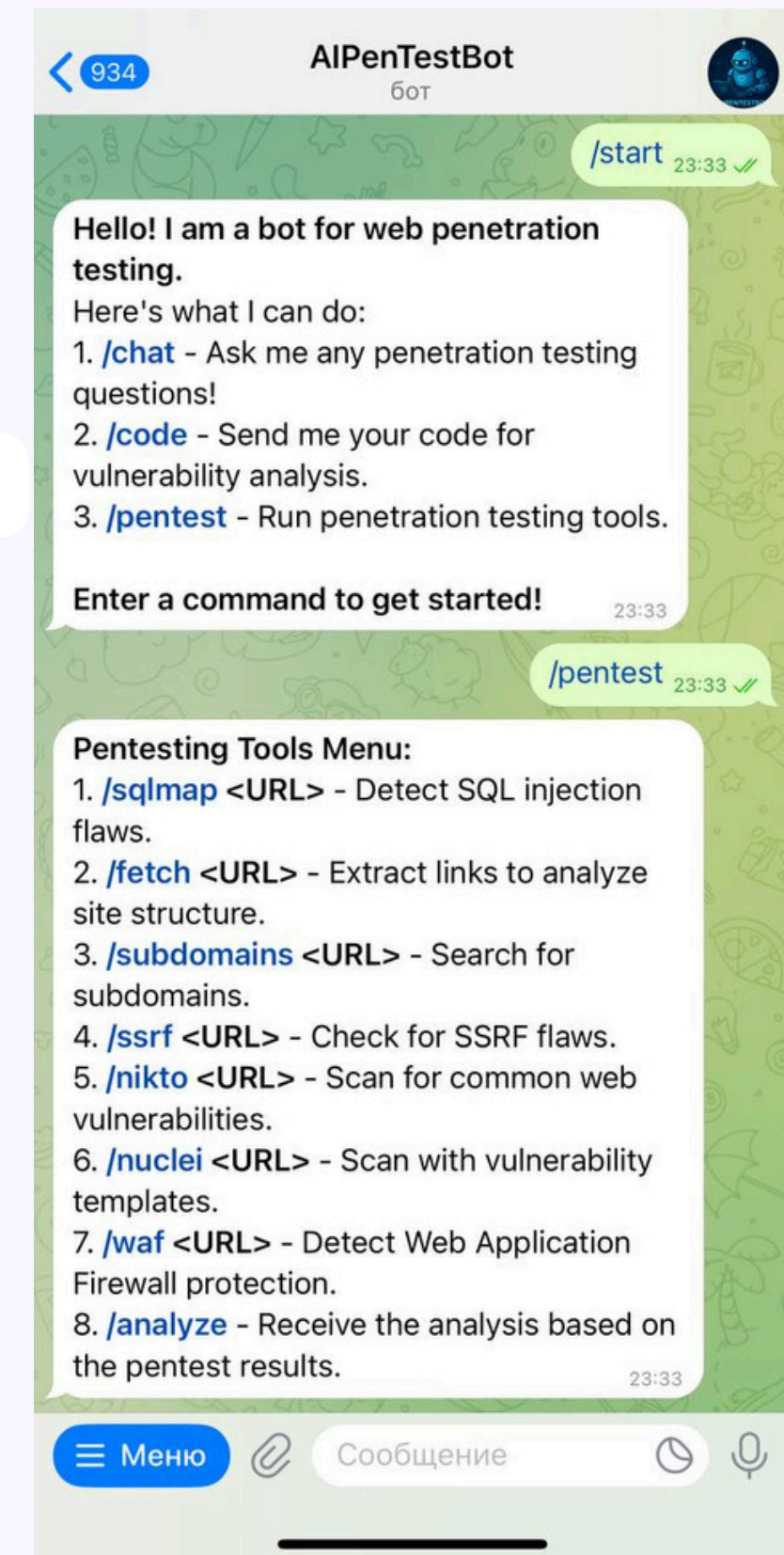
METHODOLOGY

- **Black-box testing method.**
- **Empirical method.**
- **Evaluation and testing method.**
- **Ethical Consideration method.**
- **Comparative analysis.**

ARCHITECTURE



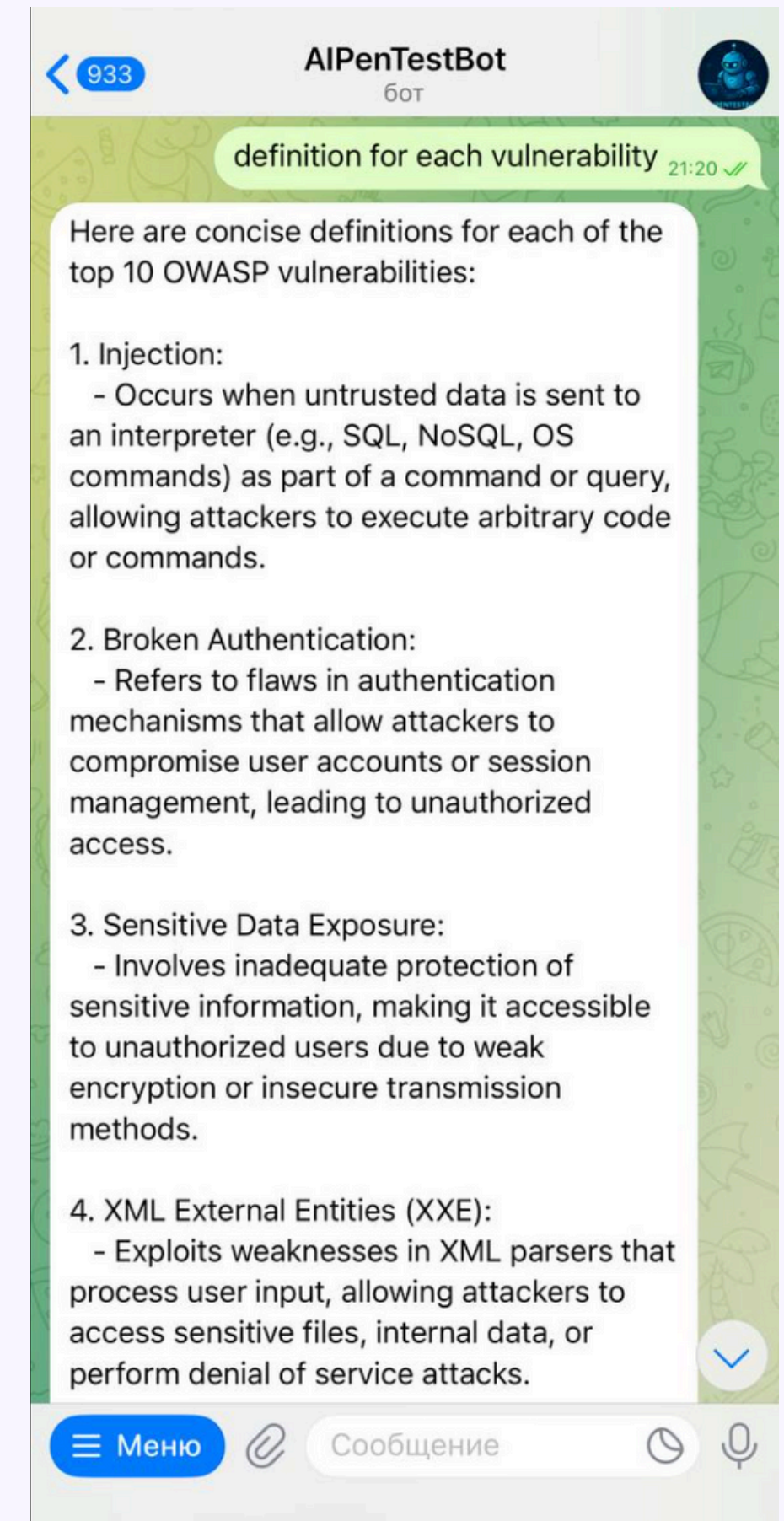
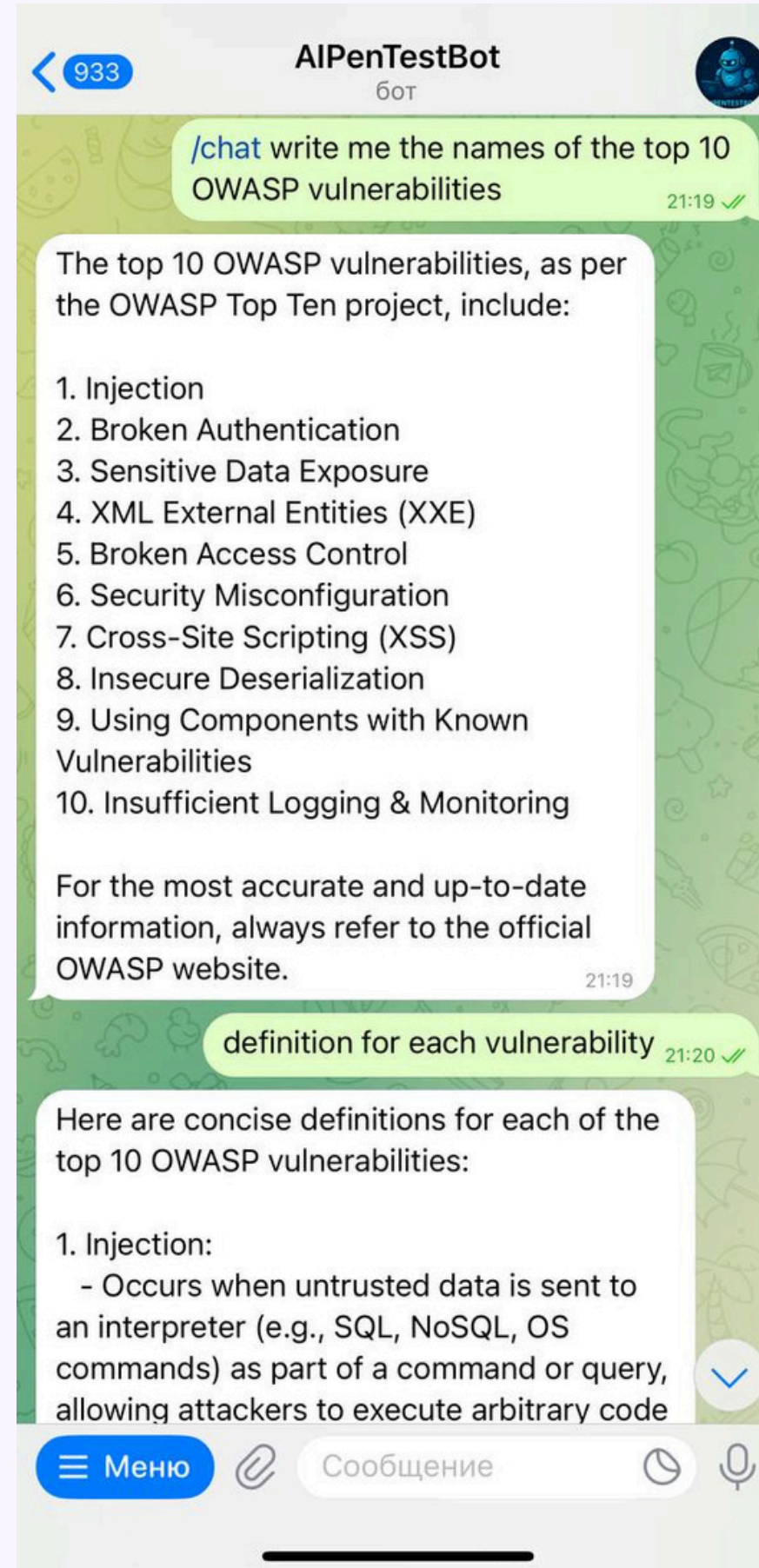
- Telegram interface
- AI-powered bot
- Pentest tools integration



CHATBOT DEVELOPMENT RESULTS: AI INTEGRATION

/chat

- Natural language input
- AI-powered security consultation
- Real-time responses
- Memory-aware interaction
- Pentesting assistant



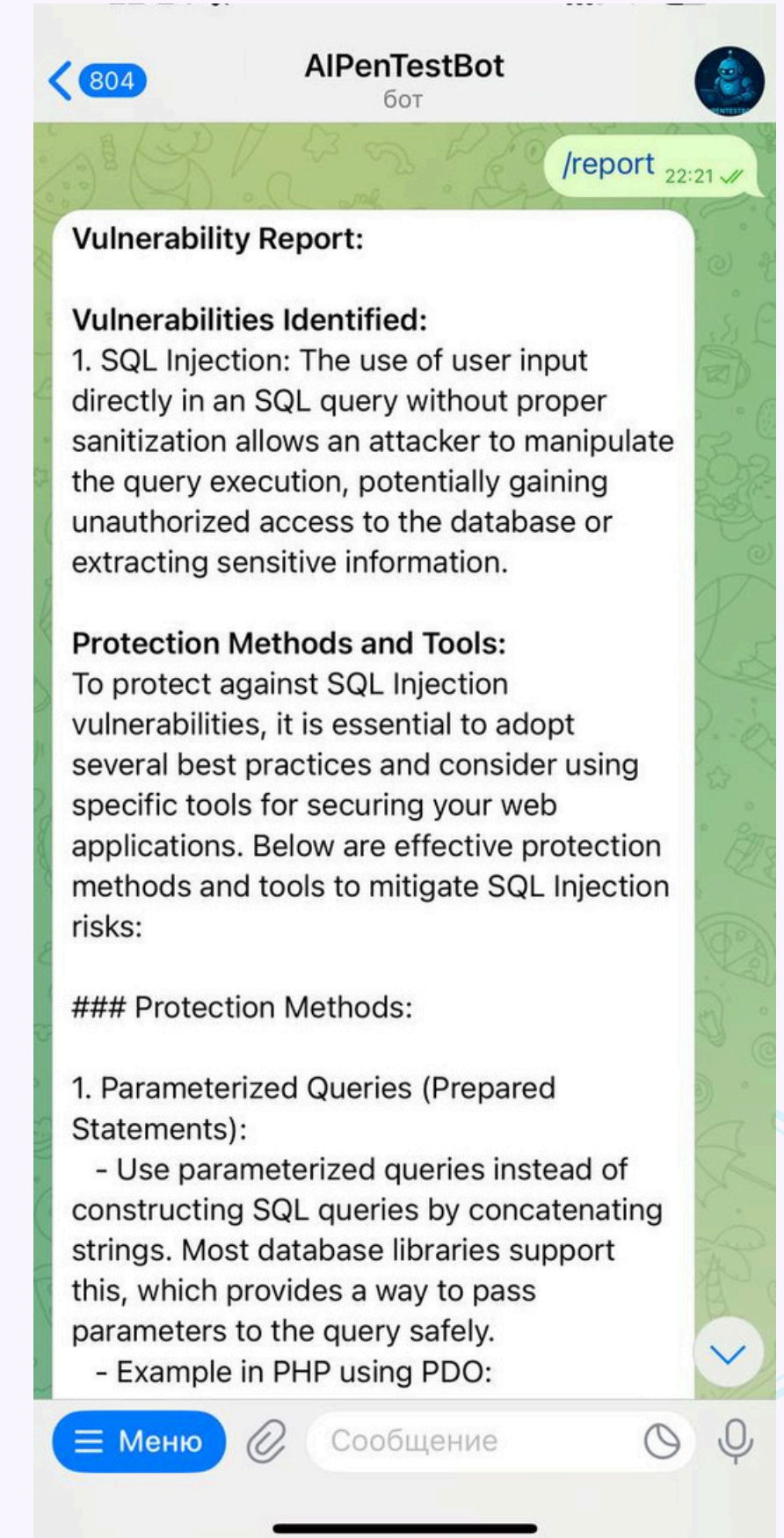
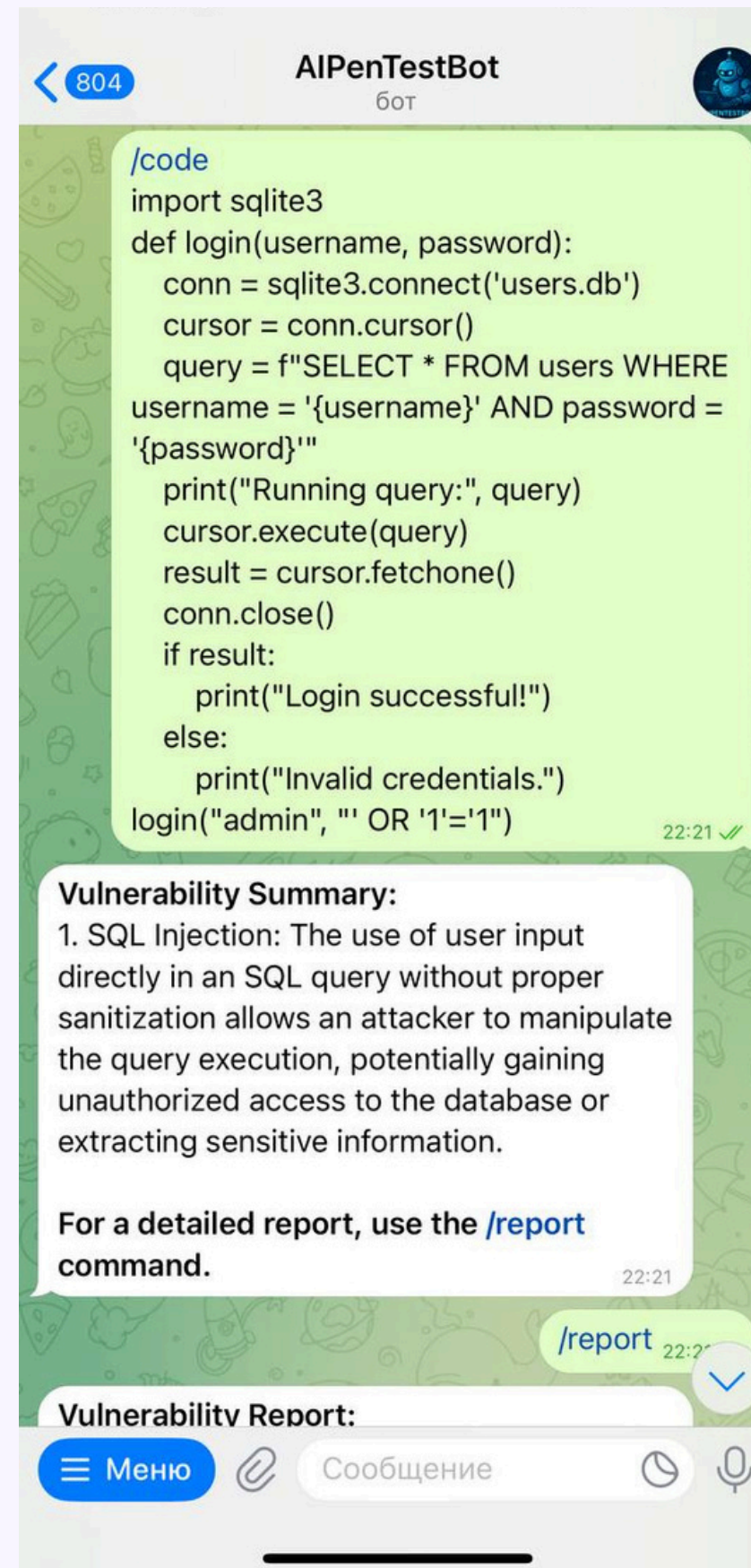
CHATBOT DEVELOPMENT RESULTS: AI INTEGRATION

/code

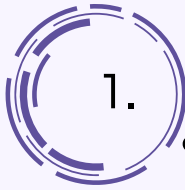
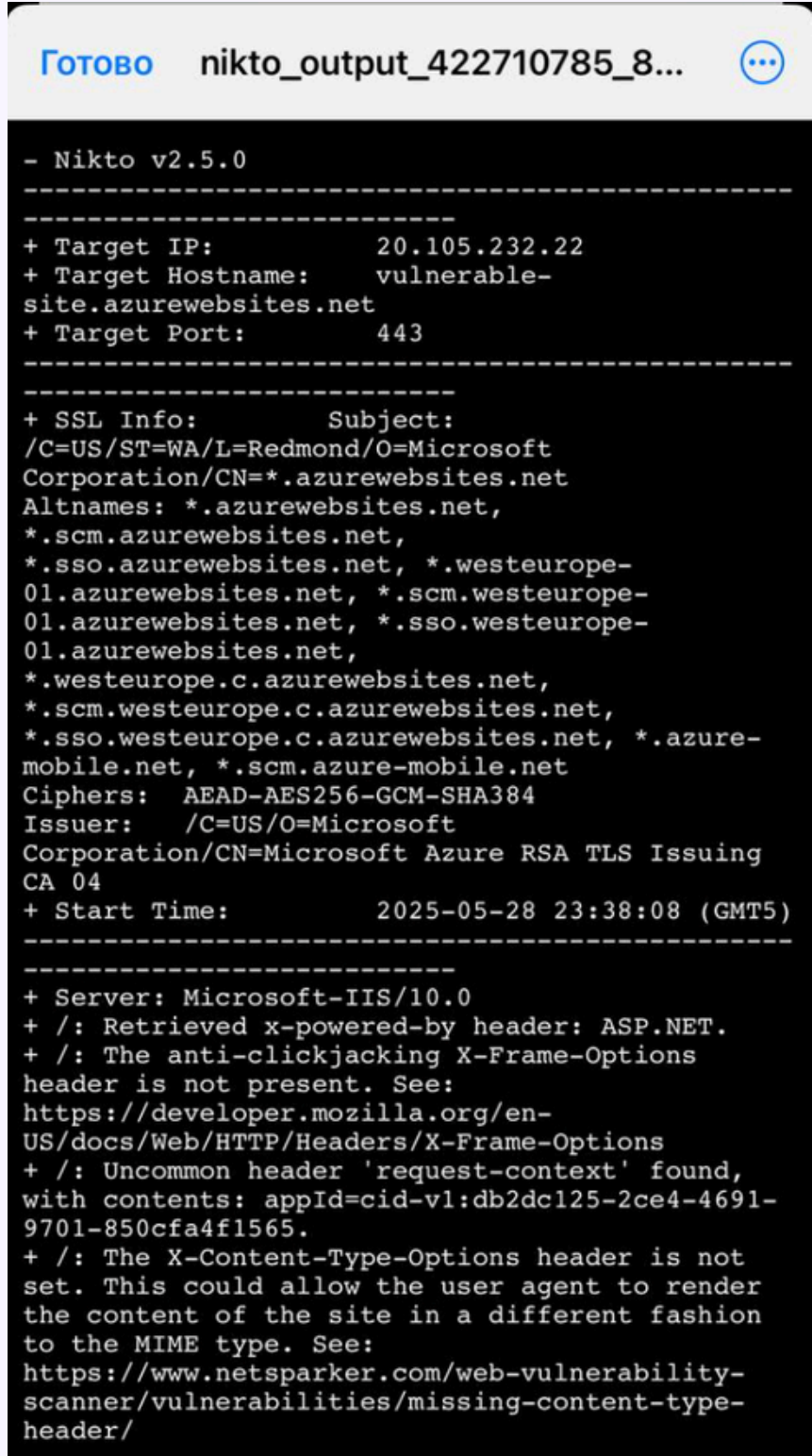
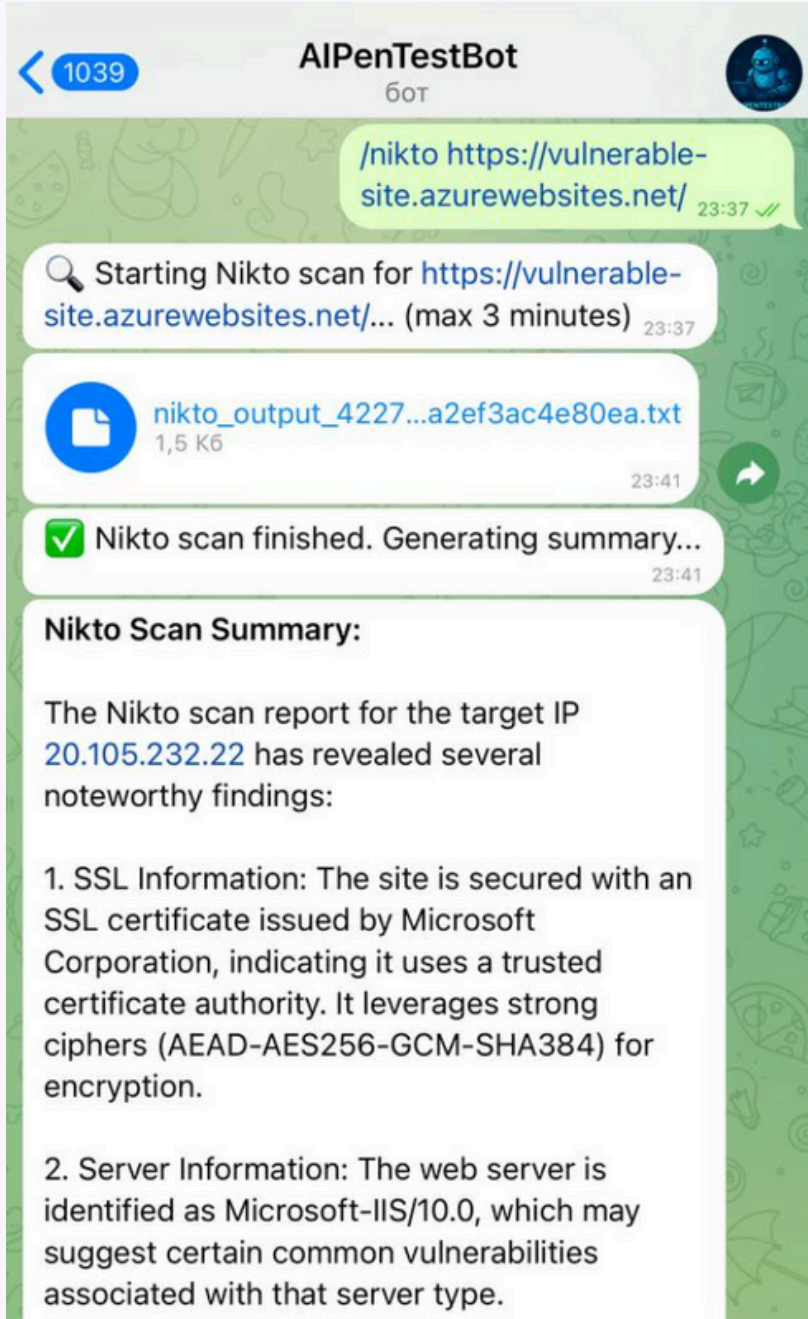
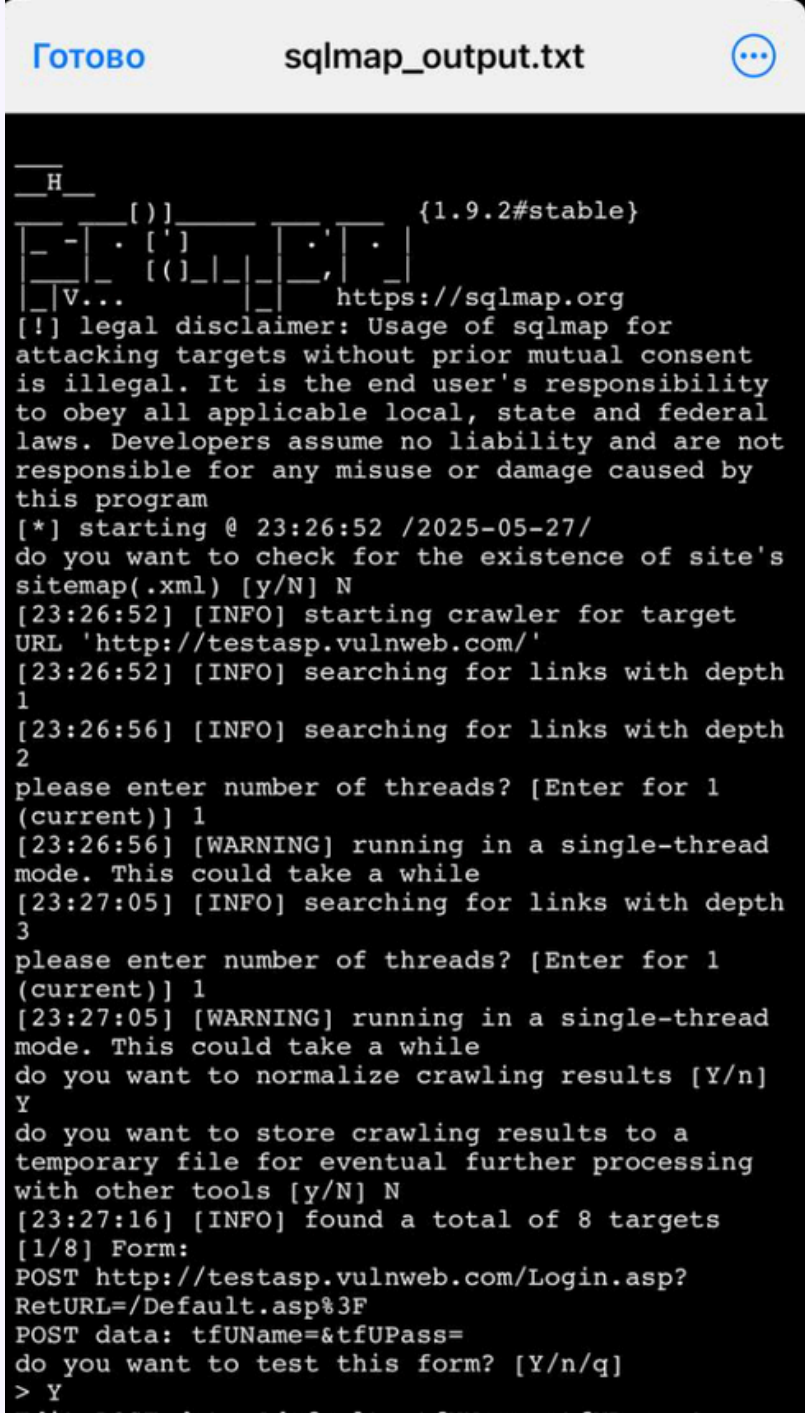
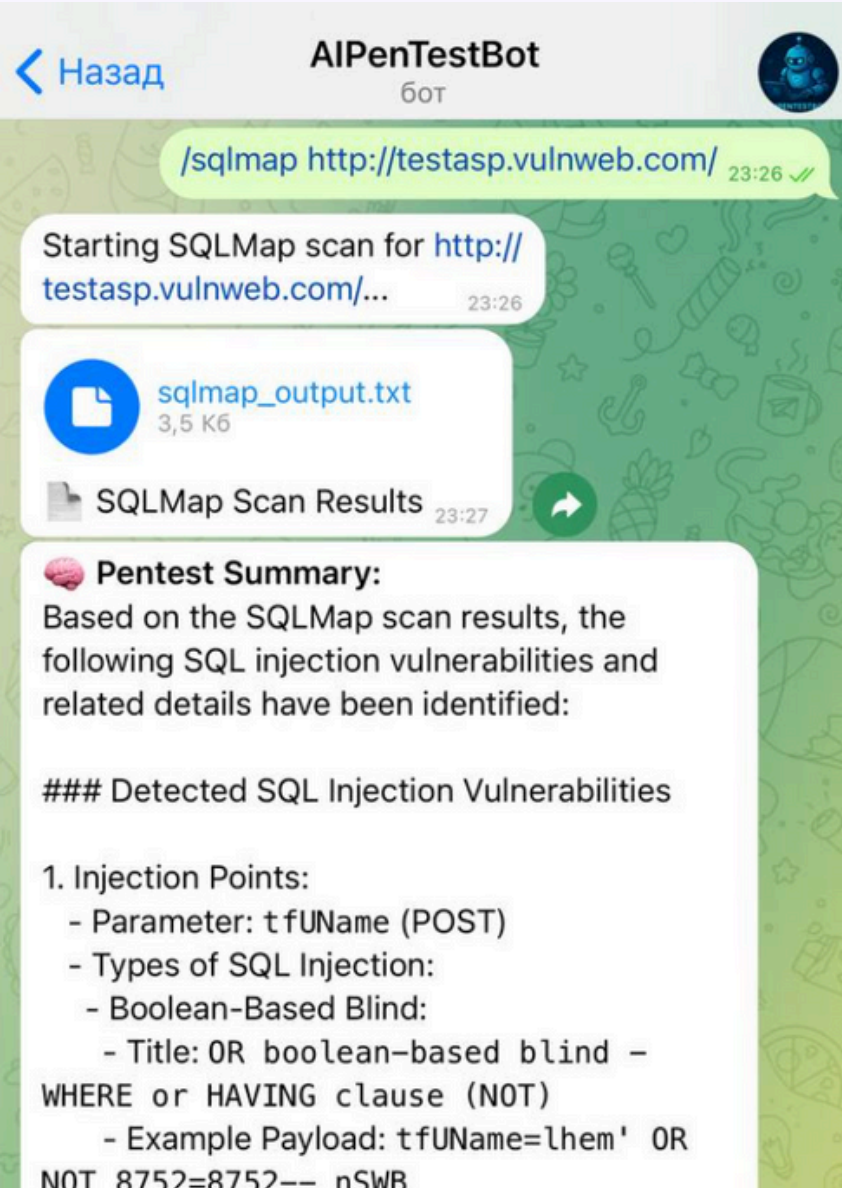
- Analyzes user code
- Detects vulnerabilities
- Instant security feedback

/report

- Generates detailed report
- Suggests protection methods
- Recommends security tools



CHATBOT DEVELOPMENT RESULTS: PENTEST TOOLS:



/SQLMAP

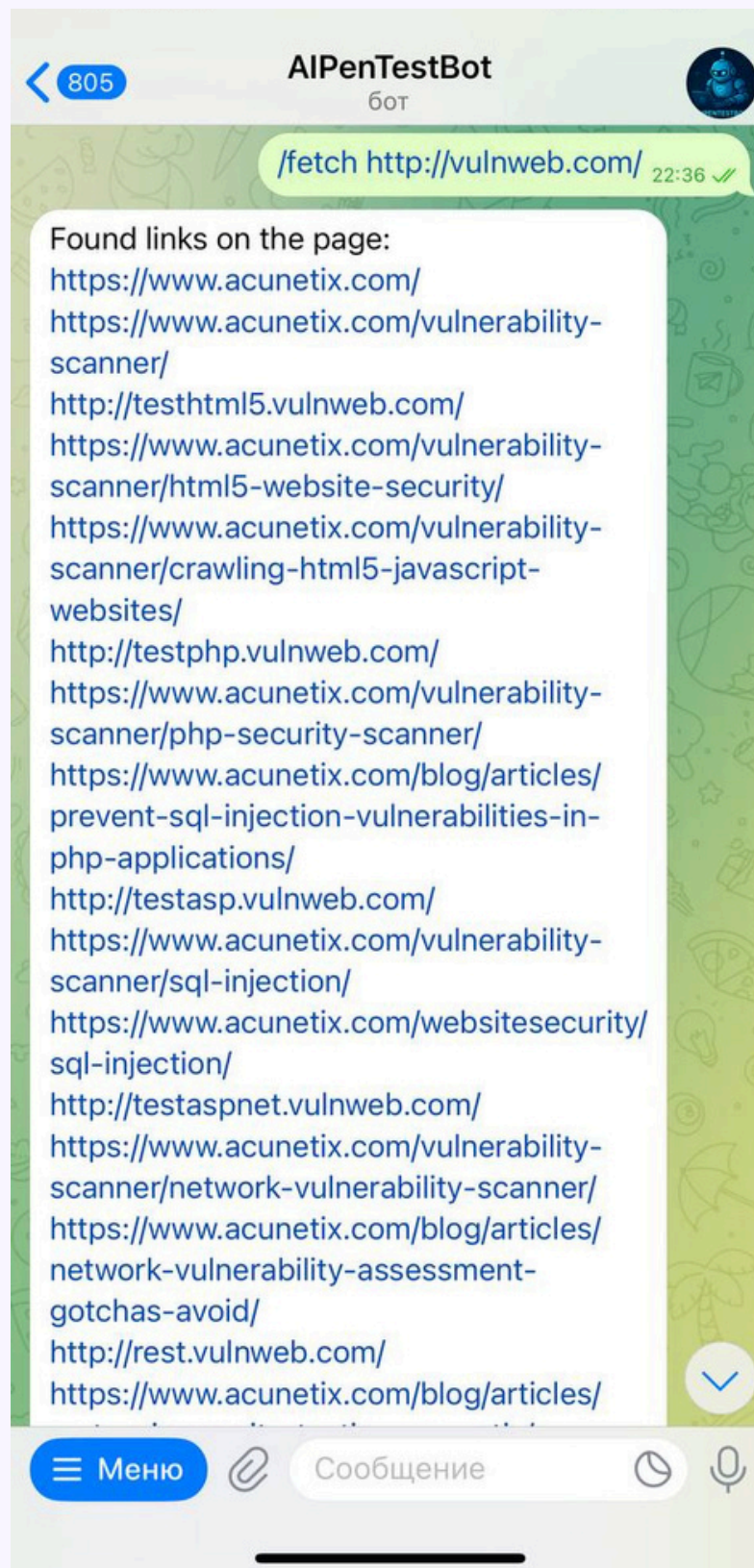
- Detects SQL injection flaws
- Analyzes request parameters
- Uses multiple injection techniques



/NIKTO

- Scans for common web vulnerabilities
- Identifies misconfigurations and outdated software
- Tests for insecure headers and cookies

CHATBOT DEVELOPMENT RESULTS: PENTEST TOOLS:



3. /FETCH

- Extracts all page links
- Maps internal structure
- Identifies potential entry points

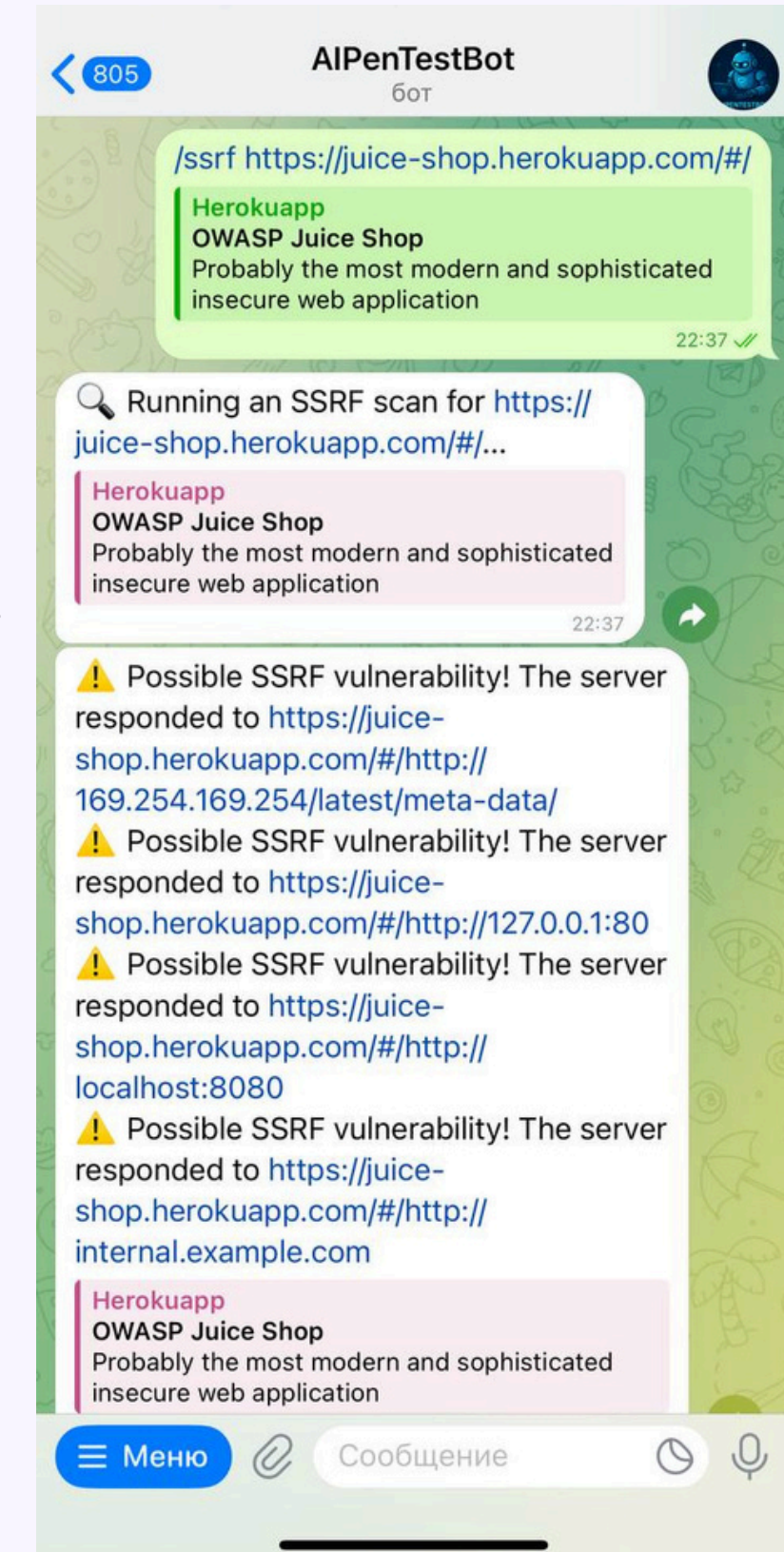
4. /SUBDOMAINS

- Discovers related subdomains
- Expands attack surface visibility
- Useful for domain enumeration



5. /SSRF

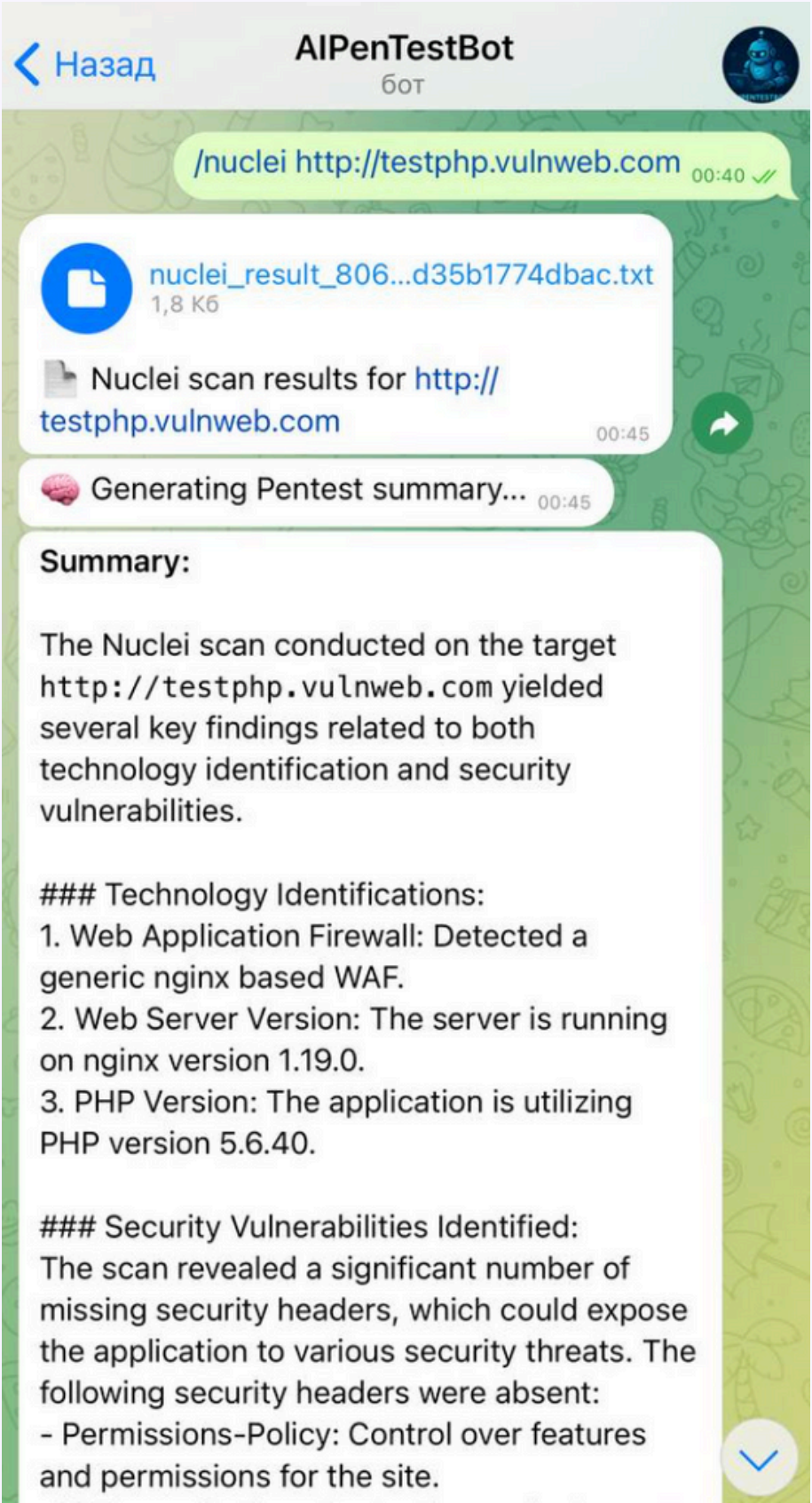
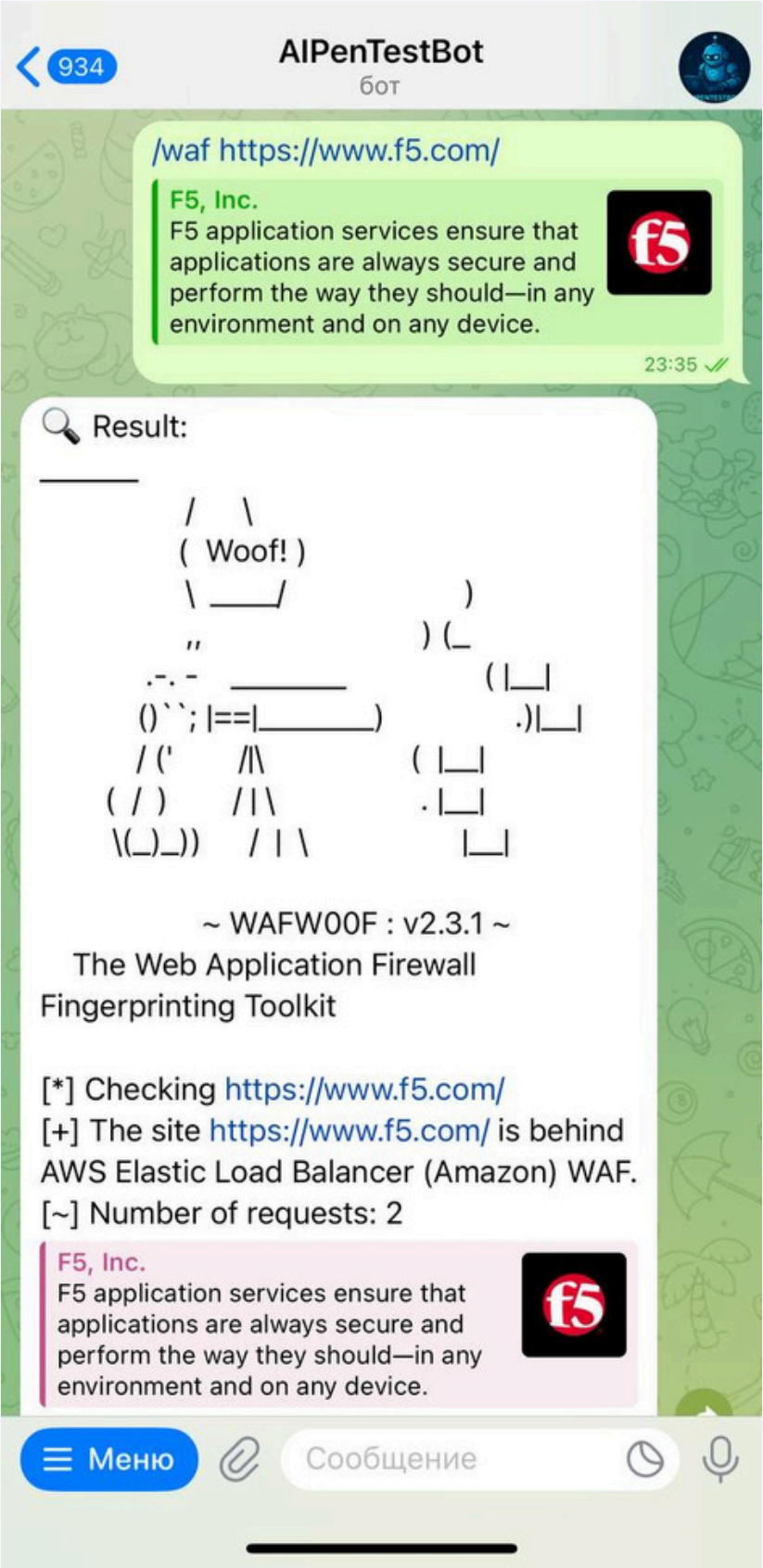
- Detects SSRF vulnerabilities
- Reveals unsafe server behaviors



CHATBOT DEVELOPMENT RESULTS: PENTEST TOOLS:

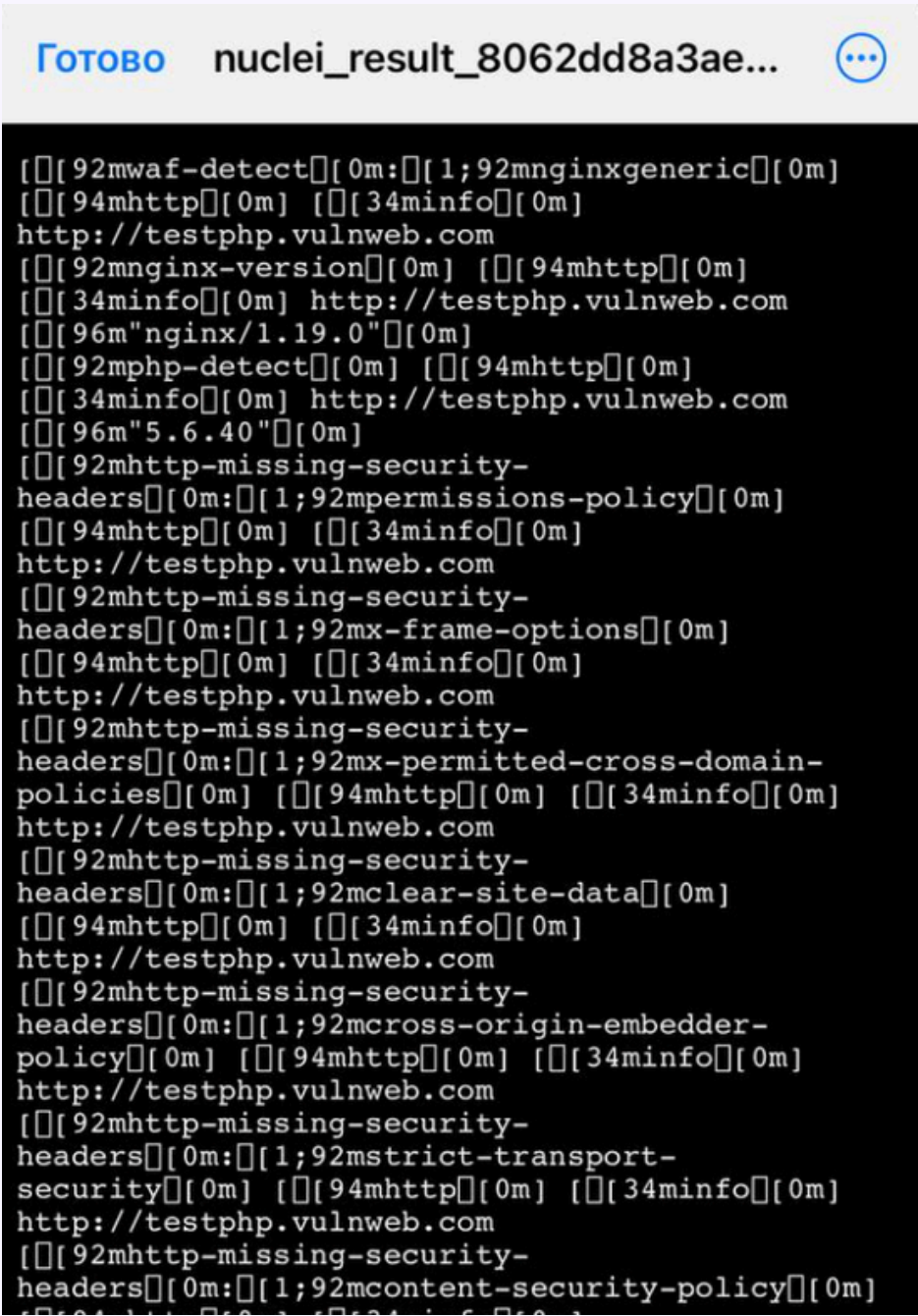
6. /WAF

- Identifies WAF vendors and rules
- Fingerprints protection mechanisms

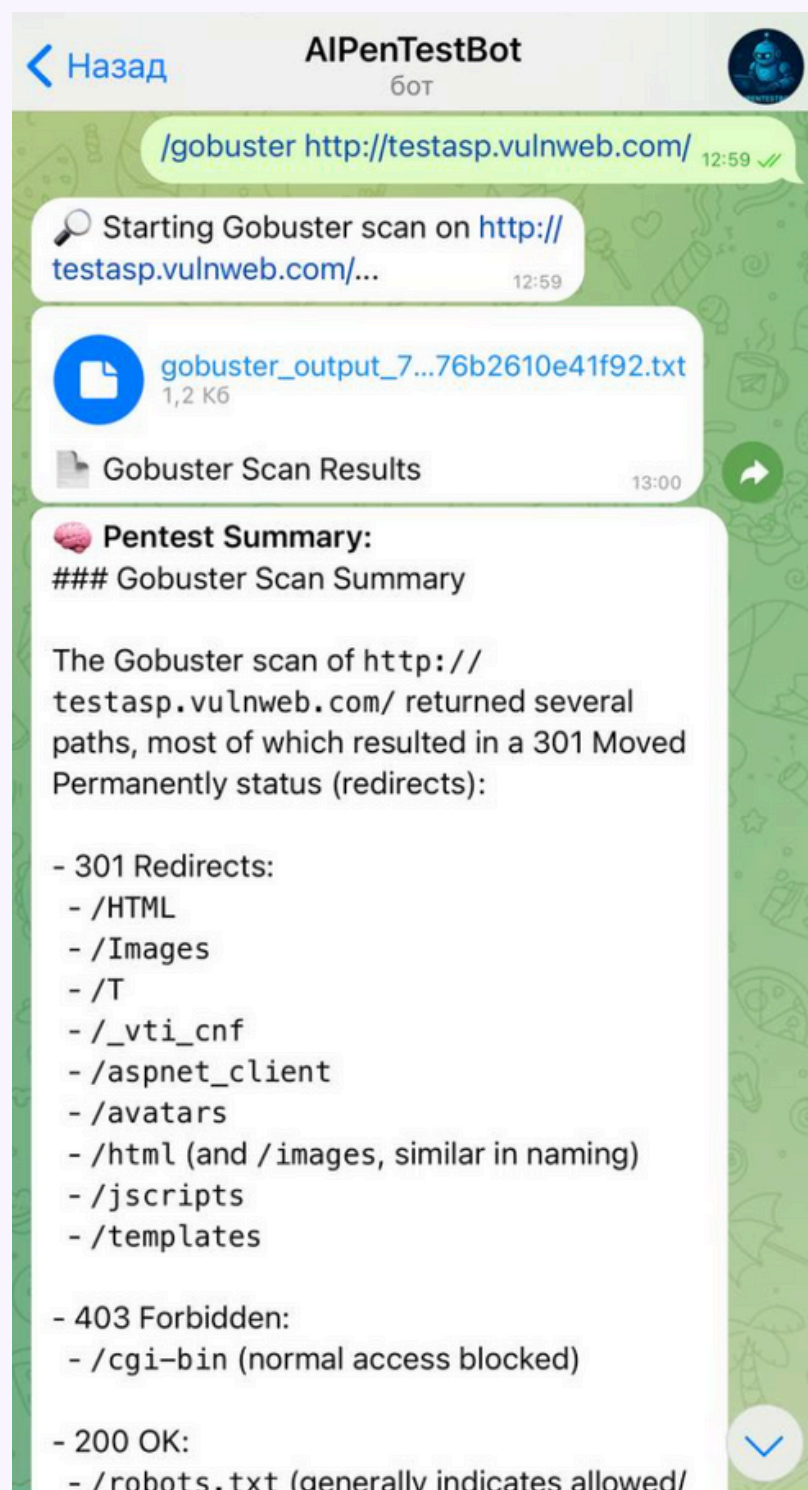


7. /NUCLEI

- Runs vulnerability templates
- Detects misconfigurations and CVEs

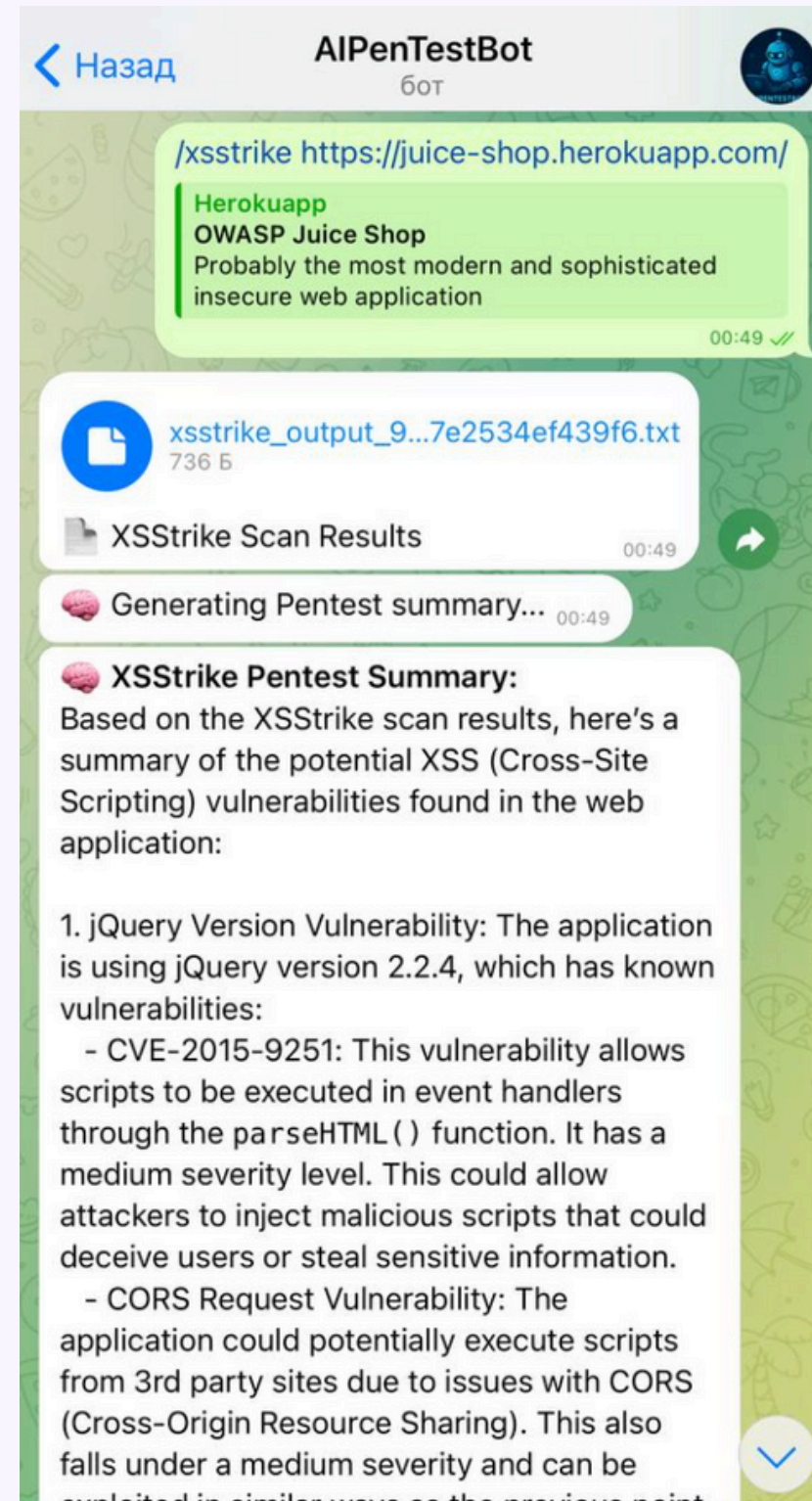
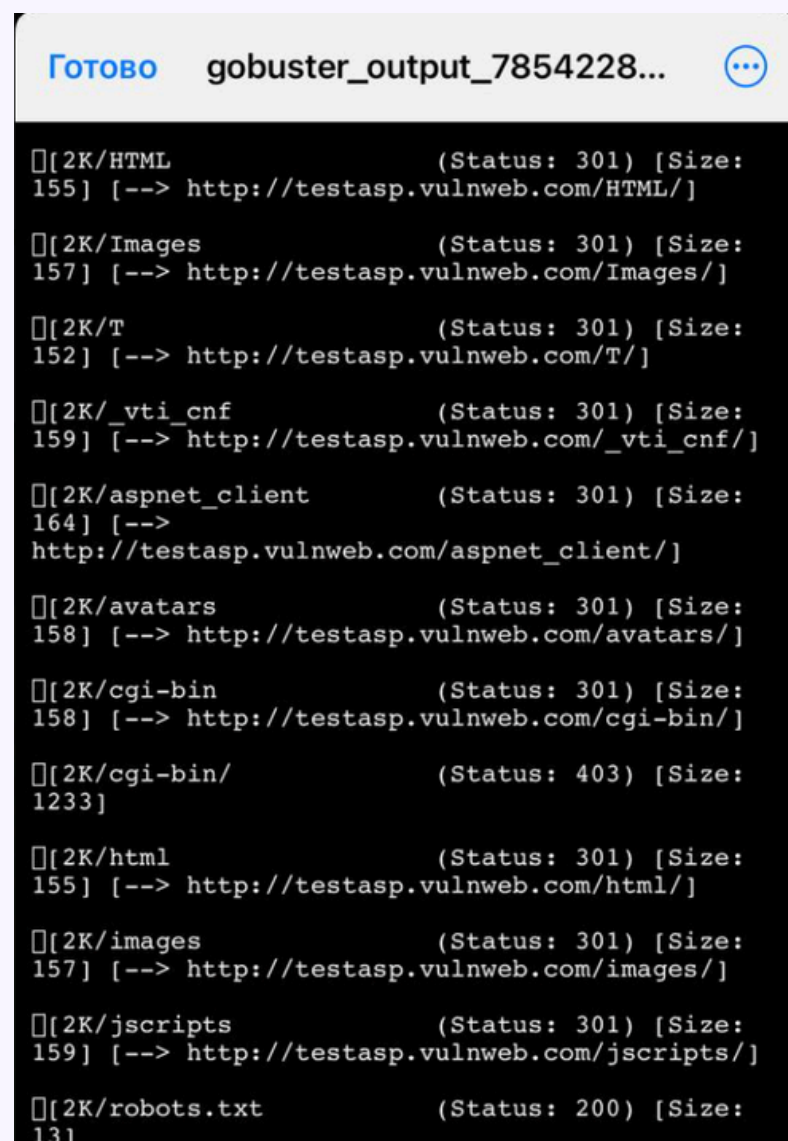


CHATBOT DEVELOPMENT RESULTS: PENTEST TOOLS:



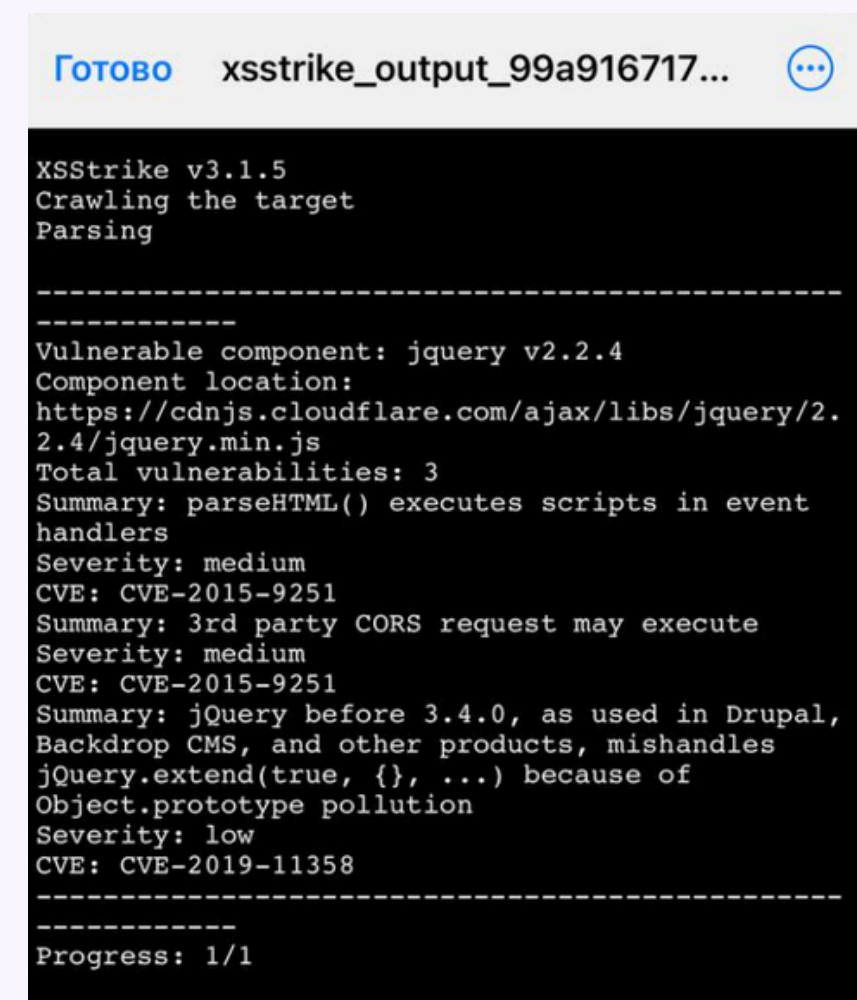
8. /GOBUSTER

- Finds hidden paths and files.
- Detects redirects and access restrictions.



9. /XSSTRIKE

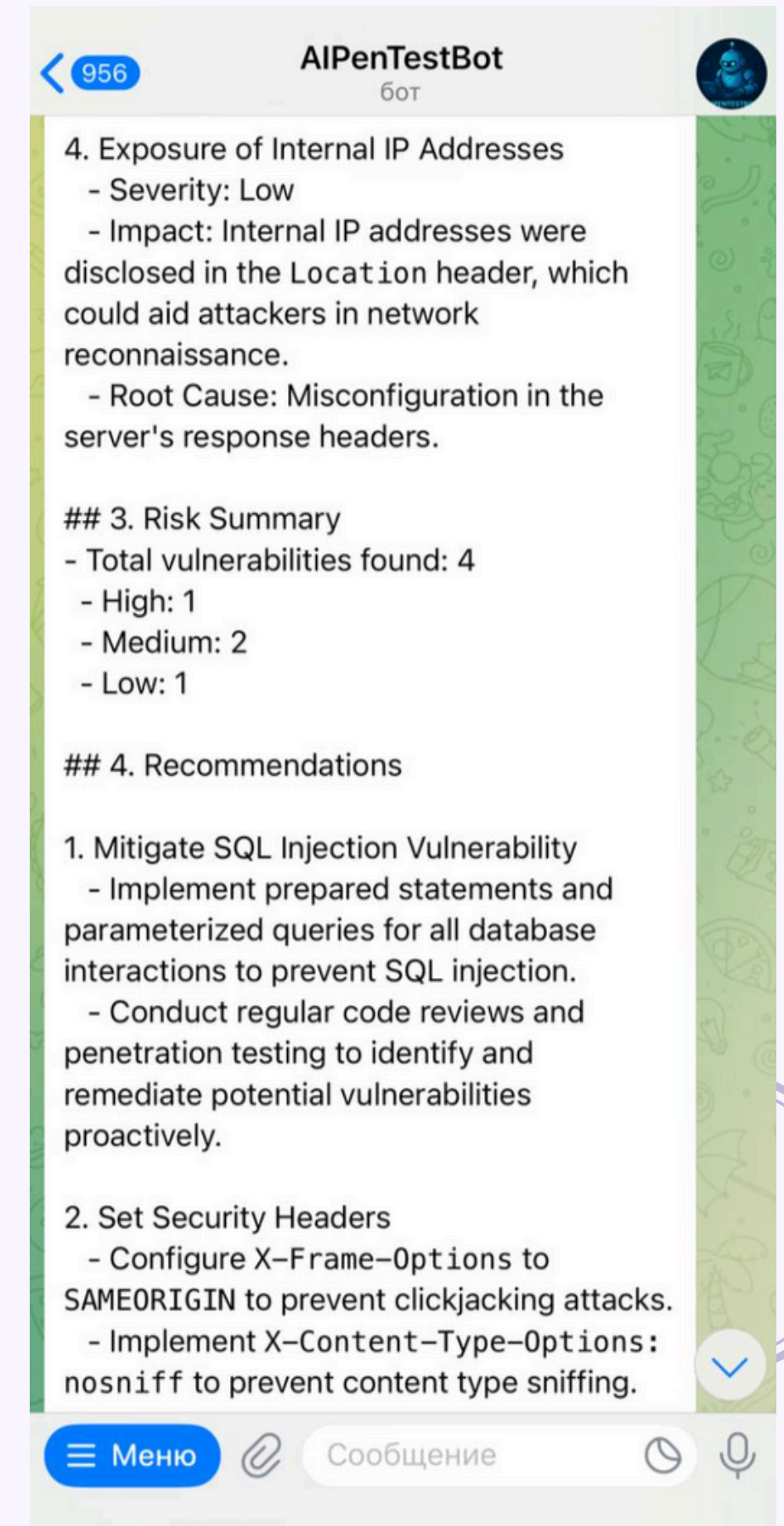
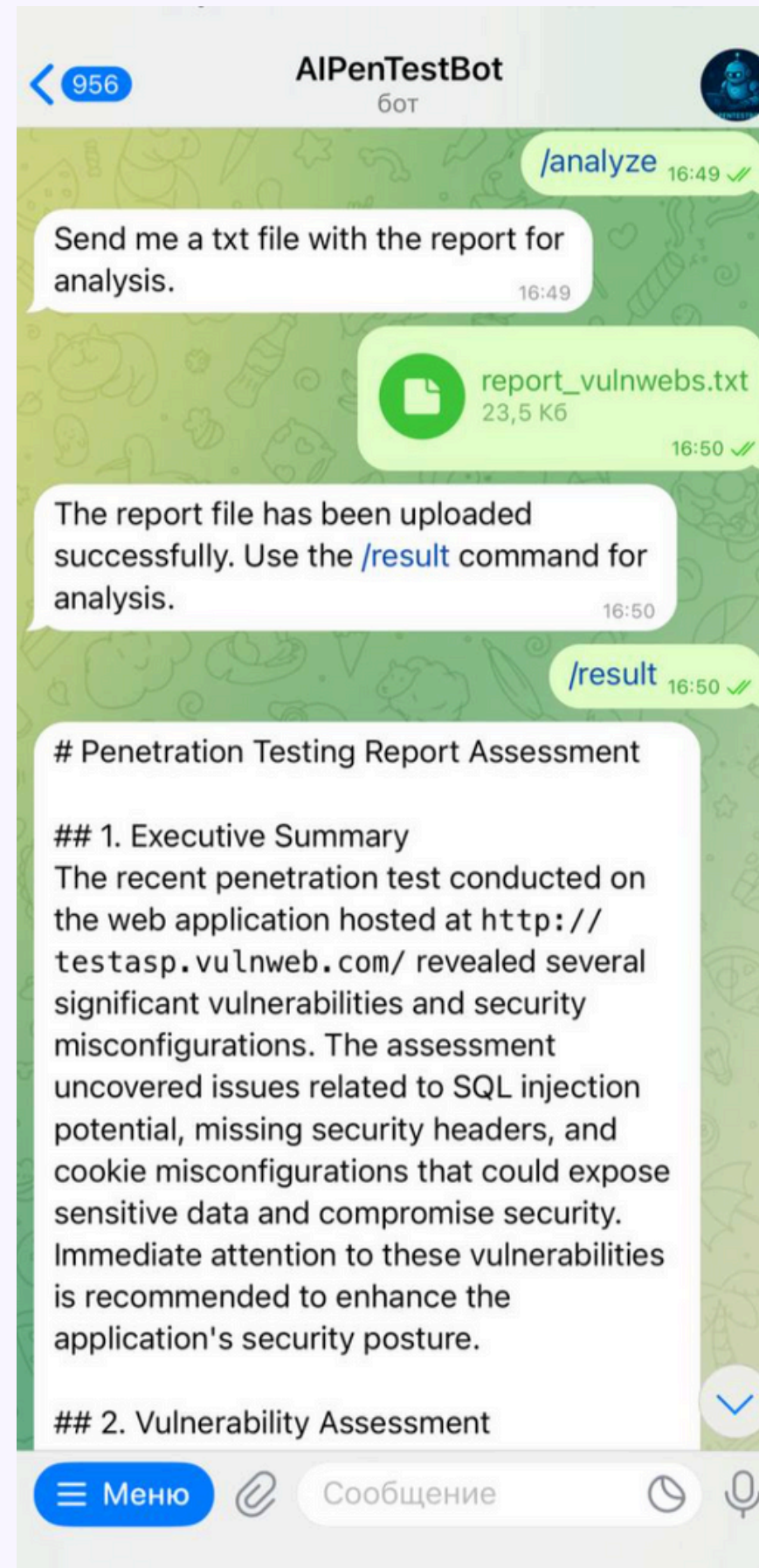
- Detects XSS vulnerabilities.
- Identifies vulnerable input points, analyzes JavaScript behavior, and related CVEs.



CHATBOT DEVELOPMENT RESULTS: SCAN SUMMARY

/ANALYZE

- Accepts .txt scan results with identifiable vulnerabilities from the chatbot or external tools.
- Summarizes detected vulnerabilities and issues
- Assigns risk levels (High/Medium/Low)
- Provides remediation recommendations and conclusion



KEY FEATURES

1.

MULTILINGUAL SUPPORT

2.

COMPATIBLE WITH DESKTOP AND MOBILE

3.

DIAGNOSTICS TOOLS

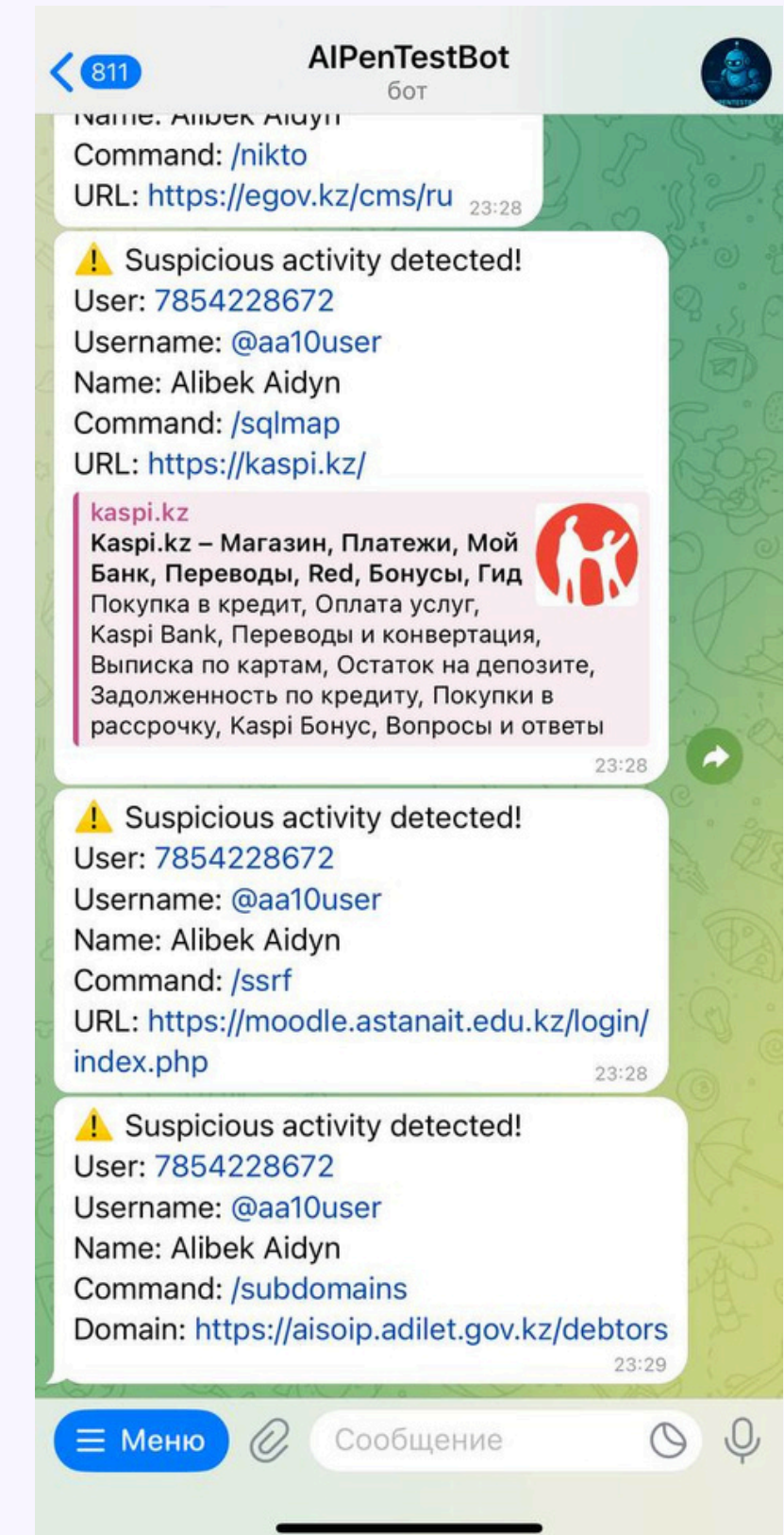
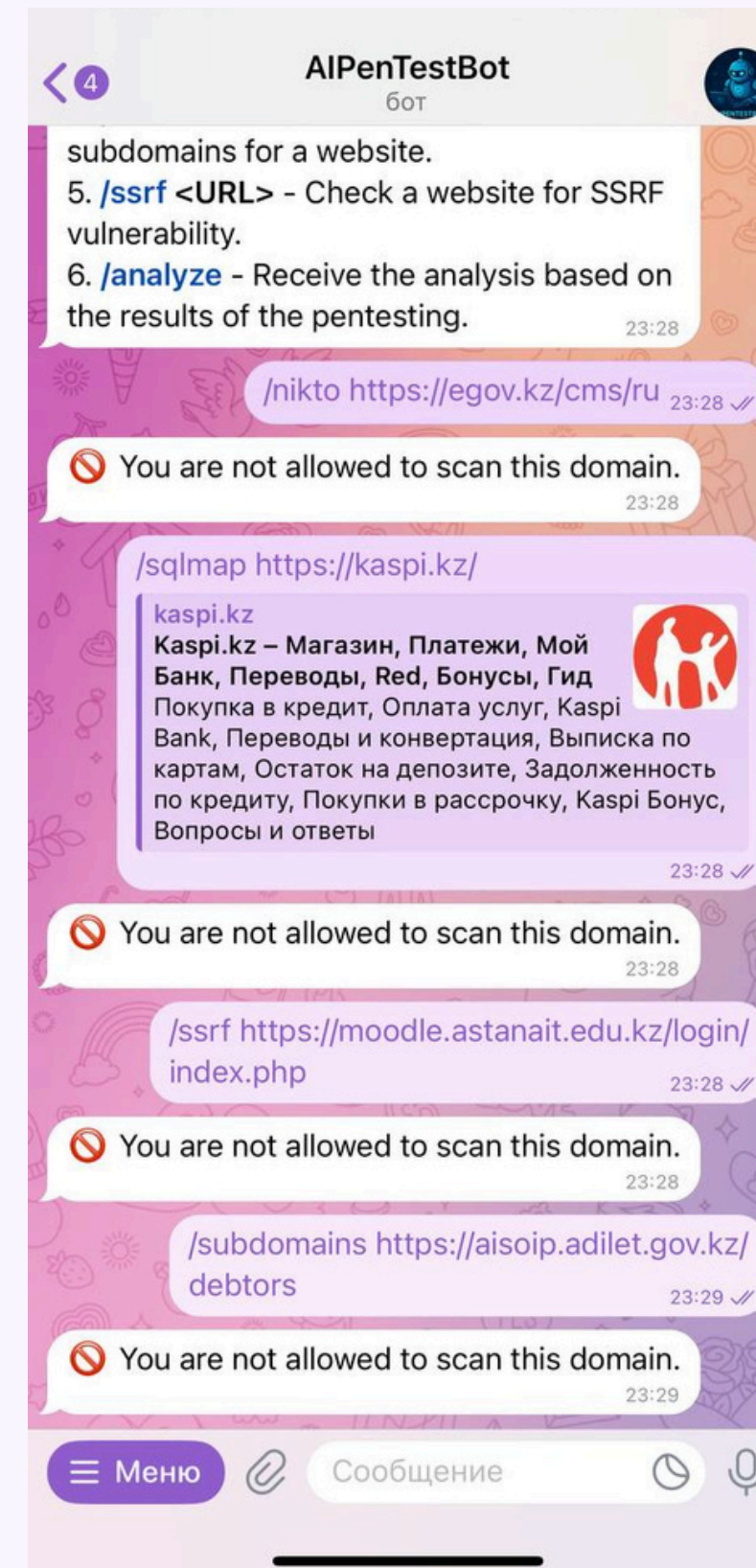
4.

REAL-TIME ANSWERS

5.

ETHICAL PRINCIPLES

- Restricts scanning of unauthorized domains
- Logs suspicious activity with user data
- Enforces ethical hacking boundaries

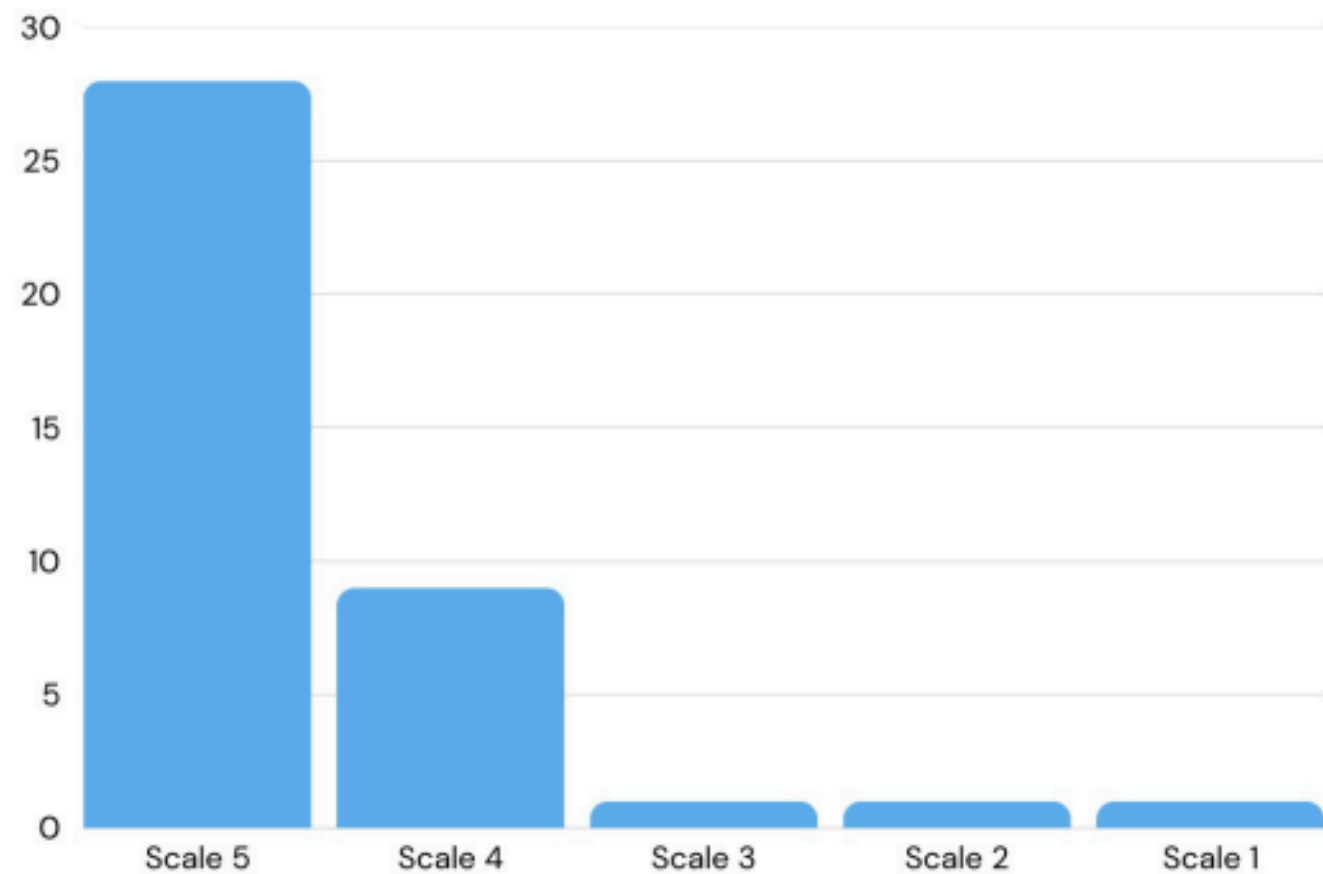


APPROBATION

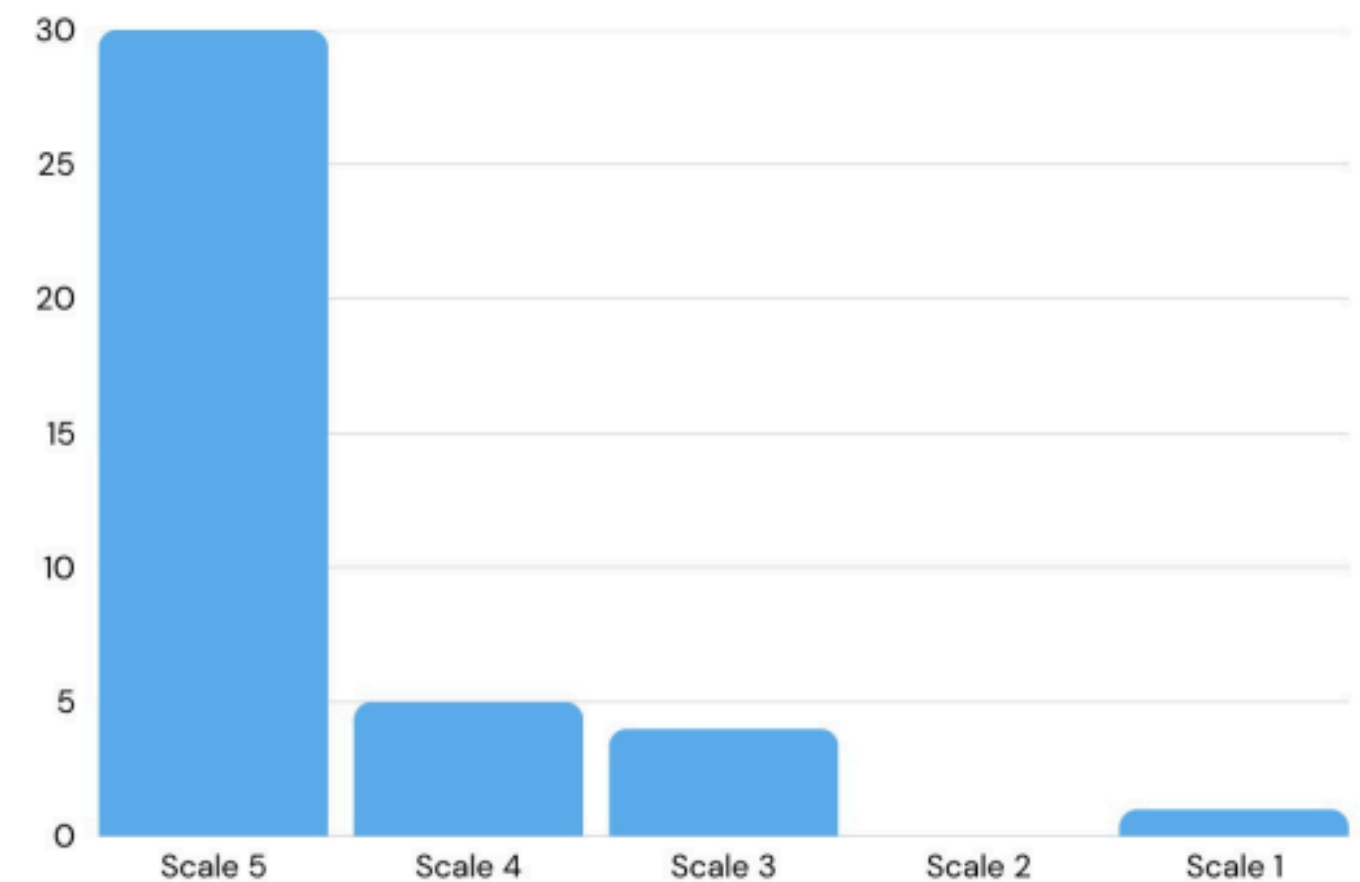
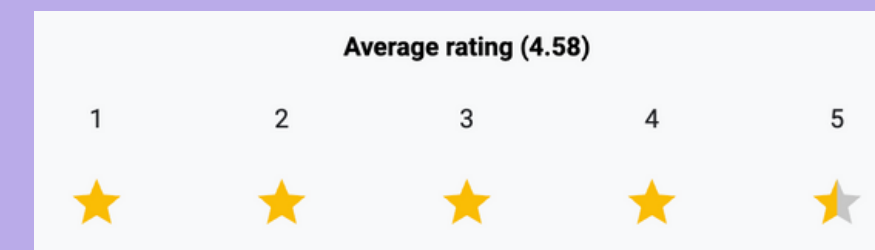
	Remote Pentest Bot	PentestGPT	AI Pentest Bot
Purpose	Remote tool execution	LLM-guided pentest with human-in-the-loop	Automated pentest; AI-based code/report analysis
Tools Used	Subfinder, Nmap, Nikto, Gobuster, Sqlmap, Hydra	None: suggests tools.	SQLMap, Nikto, Subfinder, Nuclei, WafW00f, SSRF, Fetch, Gobuster, XSSStrike
User Interface	Telegram interaction with command-based pre-defined commands	Interactive text interface (GPT-driven steps)	Telegram-based interface with command; natural language support (/chat)
AI Integration	None	Yes (Reasoning, parsing, generation modules with GPT-4)	Yes (AI-based code analysis, vulnerability summarization, and recommendations)
Output Format	Raw tool outputs sent to user	Step-by-step reasoning, task trees	Simplified responses, file uploads, detailed AI-analyzed reports
Scan Time	3-5 minute	Variable – depends on user execution	from 1 to 5-6 minutes depending on the target

SURVEY

How helpful were the scanning tools (/nikto, /sqlmap, /ssrf, etc.) for penetration testing?



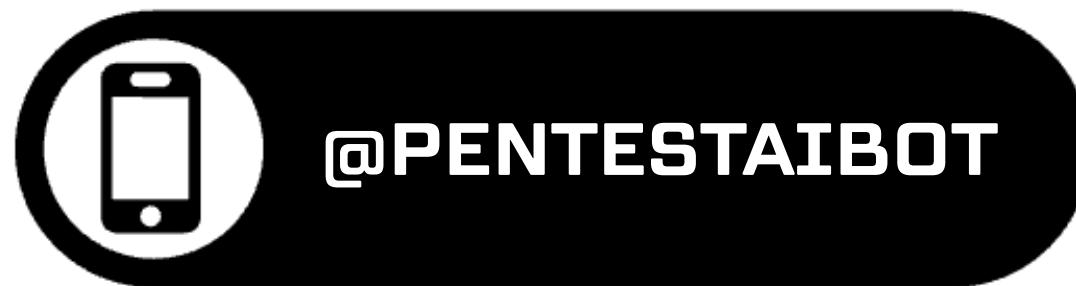
Overall, how would you rate your experience with the bot?



Conclusion

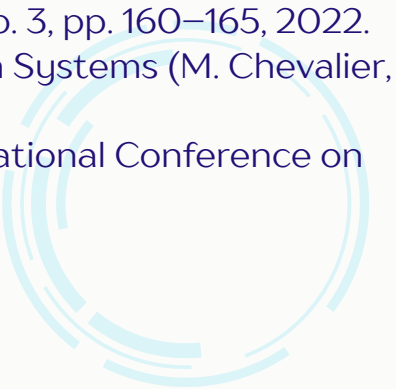
- AI-powered Telegram bot was developed to automate penetration testing tasks.
- It integrates tools like SQLMap, Nikto, Subfinder and Nuclei with a natural language interface.
- Testing showed high reliability and usability value.
- The bot simplifies security workflows and supports both novice and expert users.





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APPENDIX

A. Code listing

Here is all code with deploying on github:
<https://github.com/bagilaaa/Chatbot.git>

B. Link to video:

https://drive.google.com/drive/folders/1Zfy3rcqybbr8piol7QBPnhiz_gg_ecnB?usp=sharing